Lecture 2

Poverty and Inequality

Development Economics

9 February 2026

John Rand

KØBENHAVNS UNIVERSITET

Readings Today

- TODAY: Poverty and Inequality

- PRLB Ch.6
- Ravallion (2005)
- Ravallion (2020)
- Milanovic (2024)

- Opinion piece: Esther Duflo TED talk – Poverty. Thomas Piketty - Inequality

Main questions for today

How would you:

- Define poverty?
- Measure poverty?

Should we as development specialists focus more on:

- Inequality? Absolute or relative?
- Fairness?

POVERTY

Different Approaches to Poverty Measurement

- The Monetary Approach (Economist Approach)
  - Poverty is a shortfall in consumption (or income) from some poverty line (Ravallion)
- The Participatory Approach (Participatory Poverty Assessment, PPA)
  - Poverty is local people's perception of well-being and ill-being (World Bank)
- The Capability Approach
  - Poverty is failure to achieve certain minimal or basic capabilities, where "basic capabilities" are the ability to satisfy certain crucially important "functionings" up to certain minimally adequate levels (Sen)
- The Social Exclusion Approach
  - Poverty is a process through which individuals or groups are wholly or partially excluded from full participation in the society in which they live (Townsend)

Overview of "Measuring Poverty" – The Monetary Approach

There are three basic steps in establishing a poverty measure

1. Defining one (multiple) indicator(s) of well-being (and hence poverty).
2. Establishing a minimum acceptable standard of that indicator to separate the poor from the non-poor (The poverty line – absolute poverty).
3. Generating a summary statistic to aggregate the information from the distribution of the welfare indicator relative to the poverty line.

Economists very often prefer a simple monetary approach (In developing countries, we use consumption based on surveys).

Measuring a specific poverty line (overview)

- Traditional method to establish the poverty line use **calories** as the basis for survival as the point of the departure (**Cost-of-Basic-Needs** method – an objective, absolute poverty line).
- In short, the method proceeds as follows:
  1. Pick a nutritional requirement for good health (say, 2400 cal./day)
  2. Estimate the cost of consuming this energy requirement, using a food consumption bundle that reflects the habits of households near the nutritional requirement. This is the food poverty line z^f
  3. Add a non-food component, z^{nf}
  4. The overall poverty line is the sum $z = z^f + z^{nf}$
- BUT: Let's go into the details ☺

Measuring a specific poverty line (step 1)

As input, we must have information on

- Food consumption expenditure by food item.
  - Time-consuming
  - Measurement error
- Average shares of food expenditures, which the poor spend on each food item. E.g. 20% on maize, 15% on cassava, etc. (food basket for poor).
- Caloric contents of each food item.
- Prices of the food items.

Measuring a specific poverty line (step 2)

- Multiply food shares with caloric content to derive the caloric content of one “basket”.
- Multiply food shares with prices per unit to derive the cost of one “basket”.
- Divide caloric requirement with calories per basket to derive how many baskets one needs to meet the requirement. Multiply by price of one basket.
- Problems:
  - Are surveys precise enough to distinguish between different varieties of the same food product?
  - Different prices for seemingly similar products. Which to use?

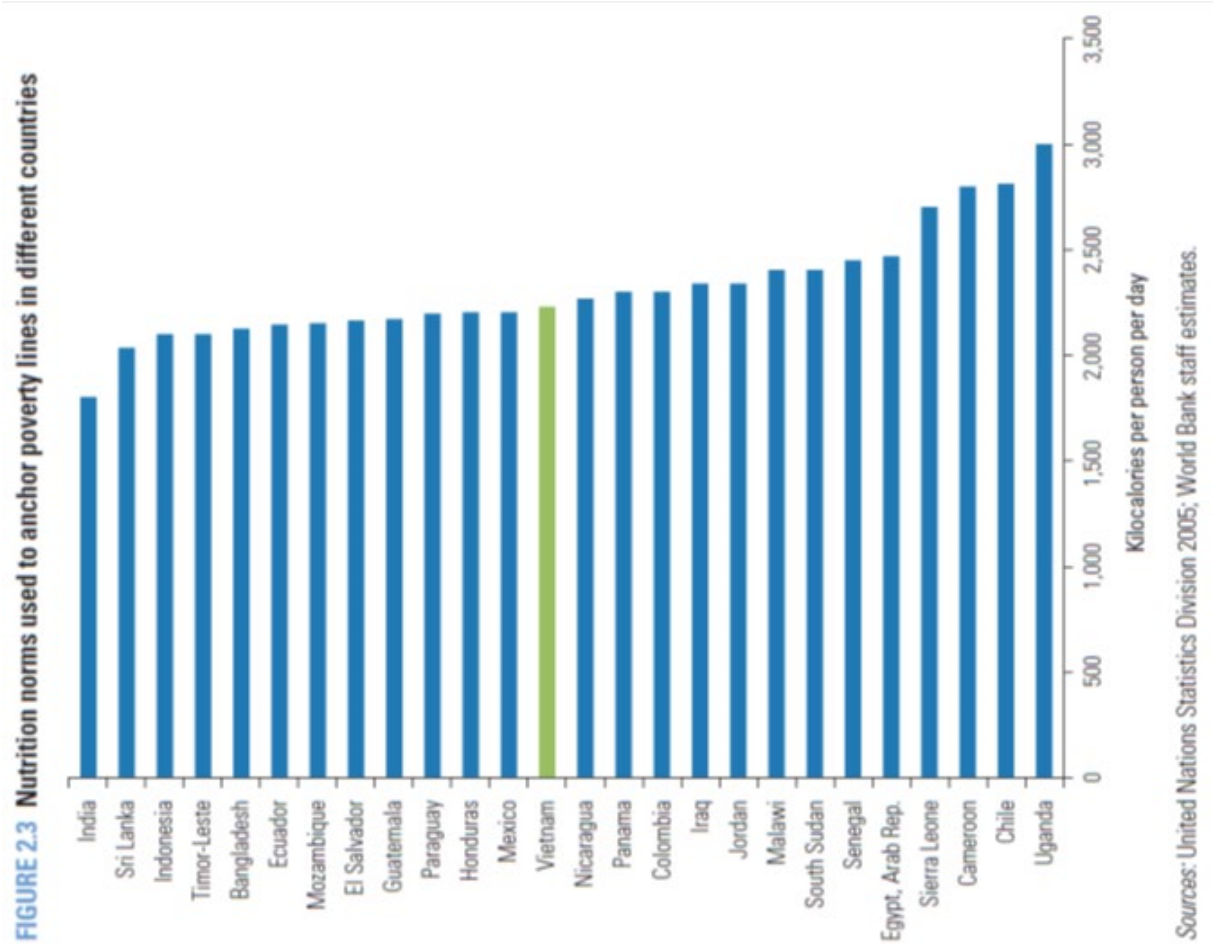
Measuring a specific poverty line (step 3)

- We now have the food poverty line. But is that it?
- Even the poorest of the poorest do not spend 100% on food and shelter.
- We define the non-food poverty line as the non-food consumption of households whose food consumption is close to the food poverty line.
- In practice, we use a weighting scheme assigning higher weight to those very close to the food poverty line.

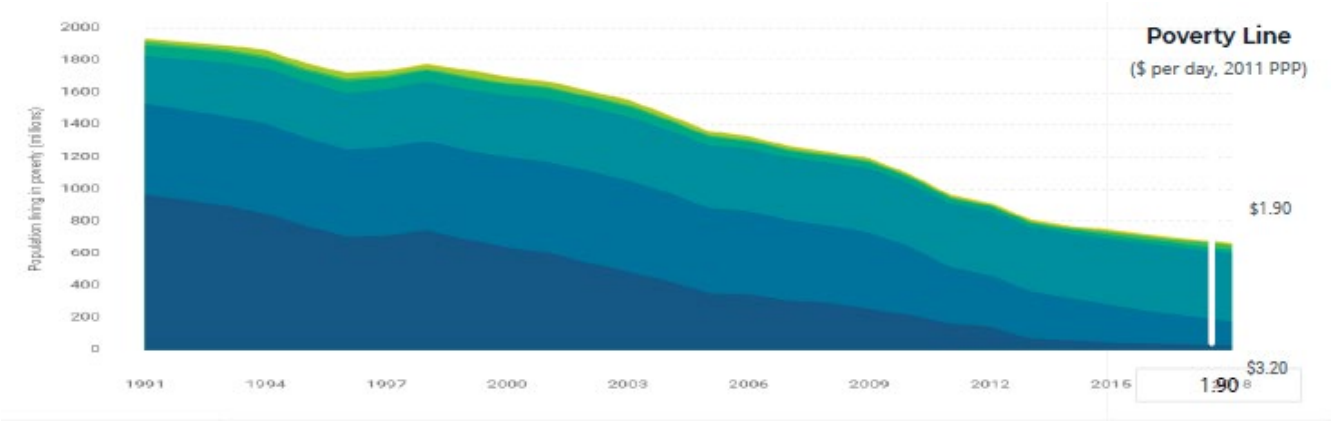
Measuring a specific poverty line (step 4)

- Add together the food poverty line and the non-food poverty line.
- General issues:
  - Different food baskets in different sub-national regions. In particular urban/rural.
  - Social effects inadequately accounted for.
  - Is it fair to have a uniform caloric requirement? Same for men, women, children, office workers, farmers, etc.? Same for all people across the world?
- Let me get your intuition:
  - In country X, they have a caloric requirement of 2200 cal./day and a poverty rate of 10%.
  - What are the implications from changing the requirement to 2400 cal./day?

Nutritional norms used to anchor poverty lines (The reference problem)

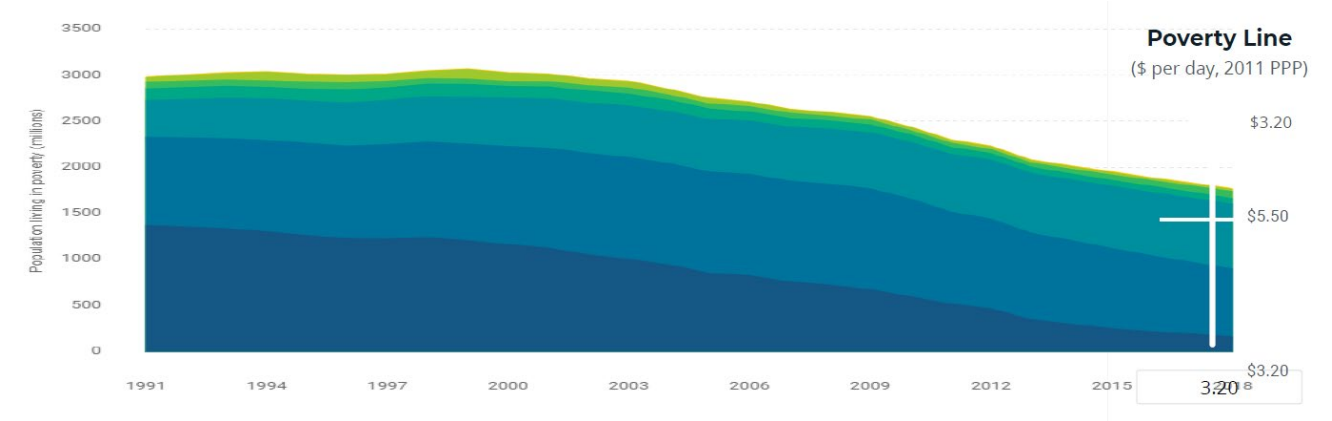


Poverty overview – different absolute poverty thresholds



1.9$ per day

659 million living below the line in 2018

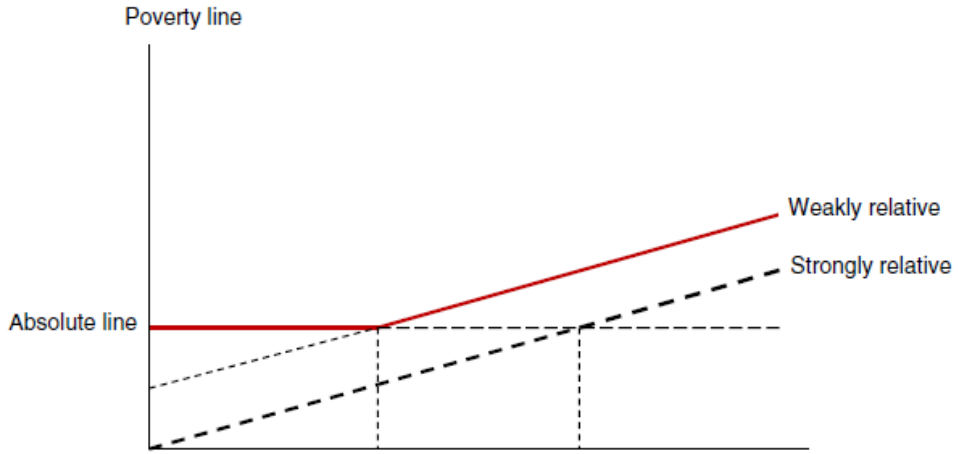


3.2$ per day

1 billion living below the line in 2018

Relative vs. Absolute Poverty

- Traditional approach: the well-being measure is usually **consumption**.
  - Why not income?
- The poverty line can be:
  - **Relative**: An expenditure level relative to the median or mean level in the population.
  - **Absolute**: A fixed level of expenditures based on (i) subjective criteria or (ii) objective criteria.
  - **Weakly relative**: Absolute until a certain threshold and relative thereafter.



Welfare may depend in part on relative incomes.
Evidence from subjective self-assessments of welfare.
May explain the Easterlin paradox (average happiness appears not to rise much with economic growth).

Strongly relative poverty lines

- Compare HH income to a poverty line that is set at a constant proportion of the current median for the country.

$$z^R = k * y(\pi_z)$$

- z^R is the relative poverty line. k is a constant. y(.) is the quantile function. π_z is a fixed percentile that defines the comparison group (often the mean or median).
- It is not clear why the quantile of any fixed percentile identifies a plausible comparison income.
  - When the poverty line is set at a constant proportion of the mean (or median), the resulting poverty measure depends solely on the distribution of relative incomes in the population.
  - If all income levels grow at the same rate, then the poverty measure will remain unchanged when the poverty line is set at a constant proportion of the mean or median.
  - Note also that national poverty lines in developing countries have an average elasticity to the mean of about 0.5, which significantly less than 1.

Weakly relative poverty lines

- Consider instead the weakly relative poverty line:

$$z^R = z^A + \beta(m - z^A)$$

where z^A is the absolute poverty line. β is a parameter. m is the mean or median.

- If $0 < \beta < 1$, then the elasticity of the poverty line with respect to m is positive but less than unity (making it weakly relative).
- But what if m < z^A? Hybrid solution:

$$z^{A+R} = \max(z^A, \alpha + \beta m) = z^A + \max(\alpha + \beta m - z^A, 0)$$

- Adding a parameter, α, that can be interpreted as the lower bound to social inclusion needs.

Empirical evidence – and new suggestion

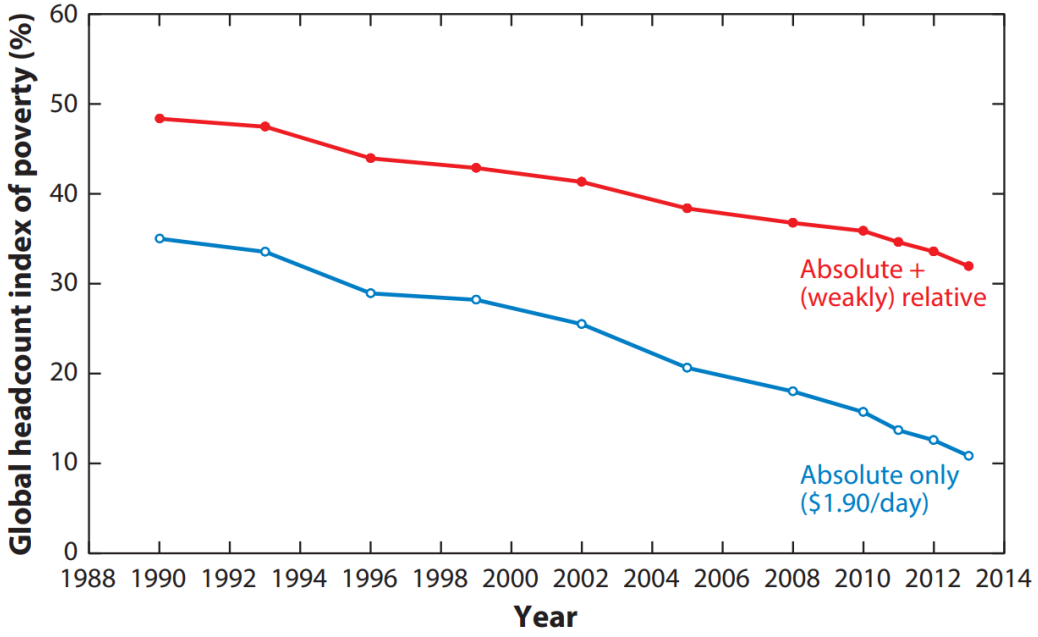
- z^A = \$1.90 a day line in 2011 prices.
- Weakly relative lines calibrated to national lines = Rank-weighted mean is the relevant comparison income, with lowest weight given to the richest.
- Implies that a Gini-discounted mean is should be used (m* = (1 – G)m), where G is the Gini index for each respective country.

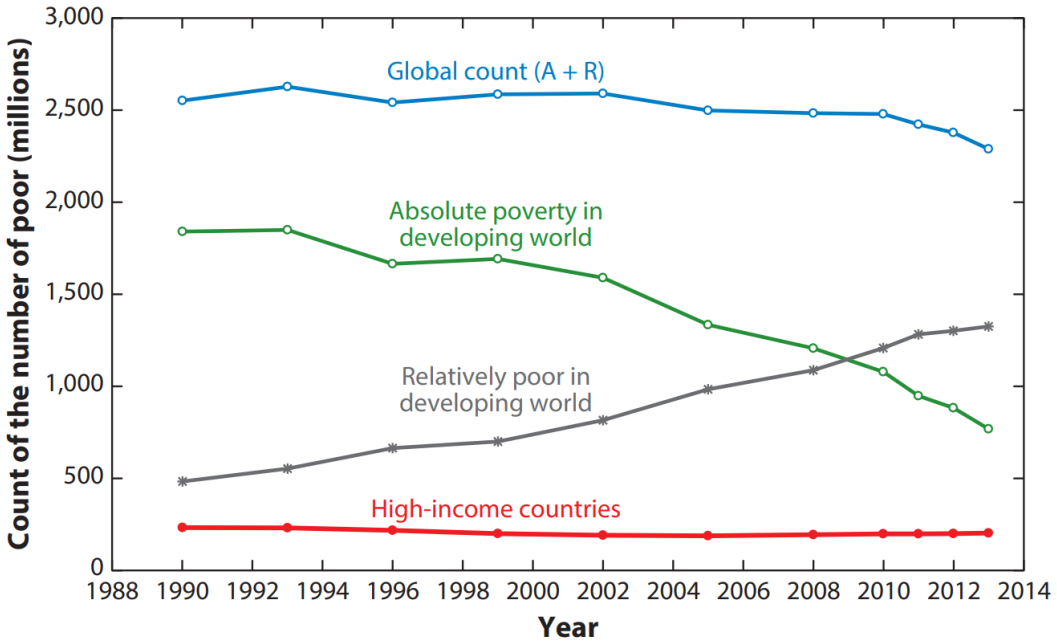
$$z_j^{A+R} = \max(\$1.90, 0.9 + 0.7(1 - G_j)m_j)$$

Why α=0.9? Why β=0.7?

- **A person is not poor globally if she is neither absolutely poor (relative to z^A) nor poor by the expected standard for the country she lives in.**

Absolute versus "new measure"





Over 90% of the poor by the A + R line are found in the developing world.
With the falling numbers of absolutely poor in the developing world, we see rising relative poverty.

Conclusion: Progress in reducing global poverty is evident. But we observe rising counts of those who are relatively poor but no longer absolutely poor. What is the correct SDG measure of poverty?

Poverty Indices in practice (1)

- **The headcount ratio (H index)**: Proportion of the population that is classified as poor.
  - Strengths: easy to understand and to calculate.
  - Limitations: insensitive either to the degree of poverty (i.e., even if every poor person's income is cut in half, *H* does not change) or to the distribution of income among the poor (i.e., even if income is transferred from a poor person to a not-so-poor person (but still considered as poor after the transfer), *H* does not change).

- **The poverty gap index (PG index)**: The extent to which individuals fall below the poverty line (the poverty gaps) as a proportion of the poverty line. The sum of these poverty gaps gives the minimum cost of eliminating poverty, if transfers were perfectly targeted.
  - Strengths: reflects depth (or intensity) of poverty; it does not imply a discontinuity ("jump") at the poverty line.
  - Limitations: insensitive to the distribution of income among the poor.

Poverty Indices in practice (2)

- **Poverty severity index (P2 index)**: Just the squared poverty gap index. By squaring each poverty gap observation, the measure puts more weight the further a poor person's observed income falls below the poverty line.
  - Strength and Limitations: Similar to the poverty gap index.

- **Watts poverty index**: The index is computed by dividing the poverty line by income, taking logs, and taking the sum over the poor. The index is distribution sensitive by virtue of its use of logarithms (transferring 100$ to a poor person counts as a larger contribution than transferring the same amount of money to a richer person).
  - Strength: The index is a distribution-sensitive poverty measure and fulfills the four axioms for a good poverty index.
  - Limitations: Difficult to understand.

Multidimensional Poverty Index (MPI)

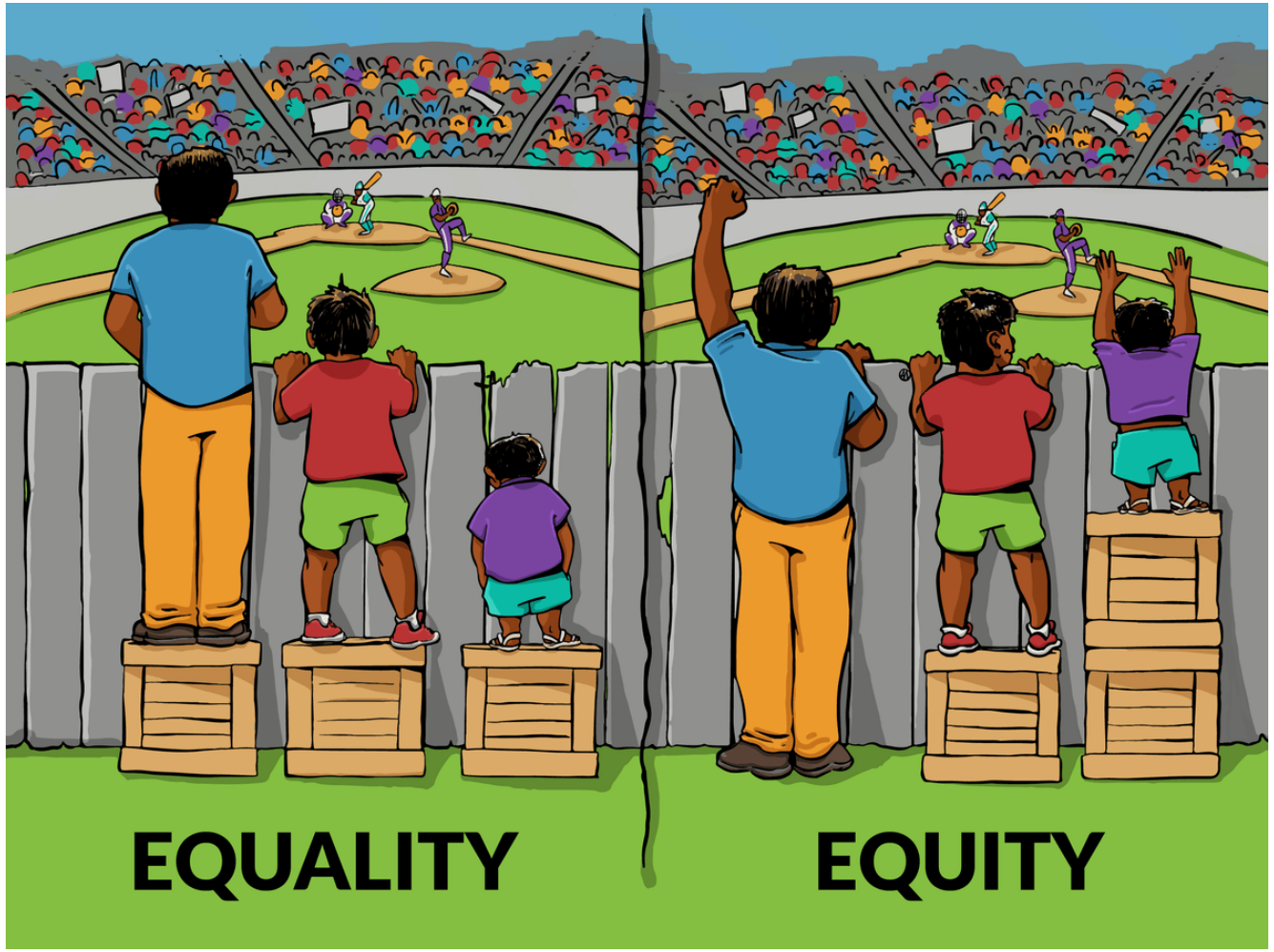
- Acknowledging the disadvantages of traditional poverty lines, researchers have increasingly turned towards MPIs.

- Essentially, advocates of MPIs argue that **we need to understand what it means to be poor**. Earning a specific amount of money or eating a specific amount of food is not precise enough.

- Key questions are "which dimensions are more important than others?" and "what are the cutoff points for being poor in specific dimensions?"
  - A lot of arbitrariness is involved – How big a problem is this?

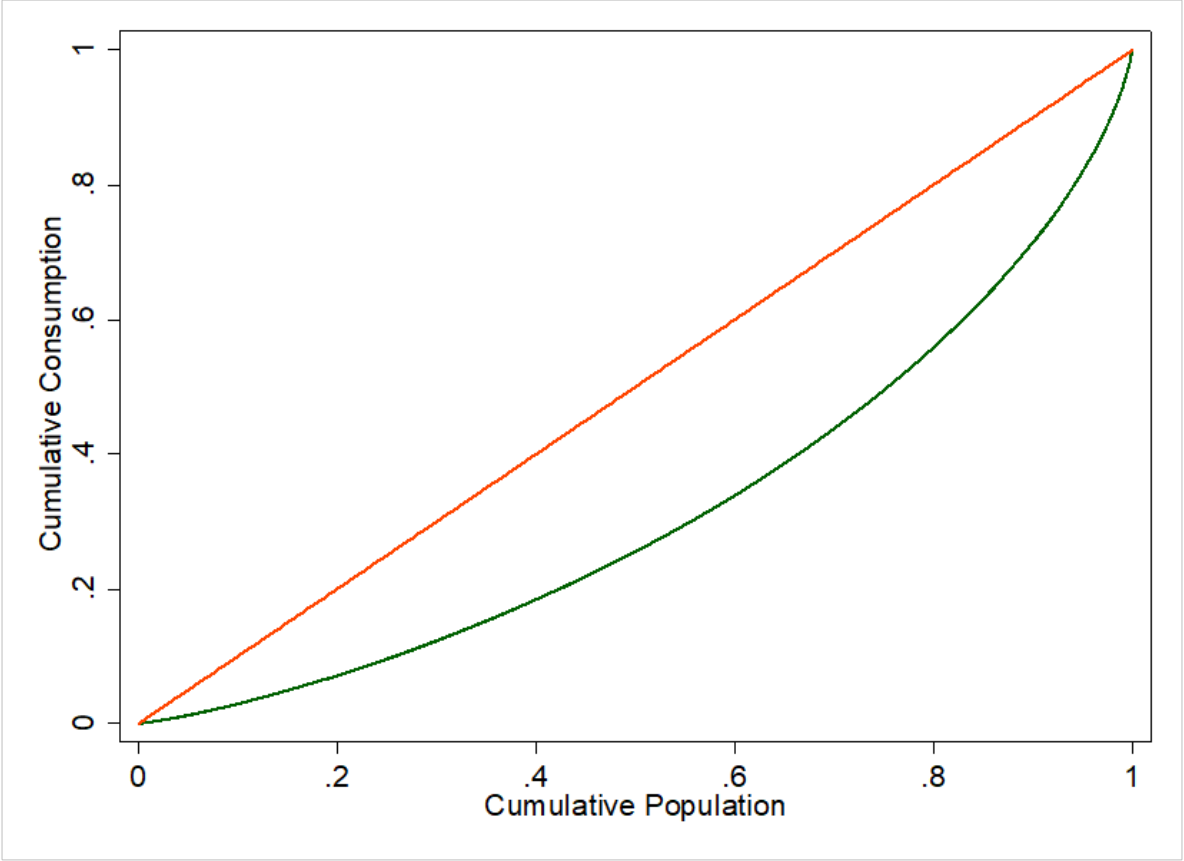
INEQUALITY

Equality "sameness" versus Equity "fairness"



Measuring inequality - the Lorenz Curve

Decile group	Consumption share, %	Cumulated share, %
1	2.90	2.90
2	4.25	7.15
3	5.20	12.34
4	6.11	18.45
5	7.10	25.56
6	8.32	33.87
7	9.93	43.80
8	12.05	55.85
9	15.61	71.45
10	28.55	100.00



The Gini coefficient and the Lorenz curve

The Gini coefficient: A/(A+B)

- "Measures the distance from perfect equality (the 45°-line)
- Defined on 0 to (n-1)/n -> 1

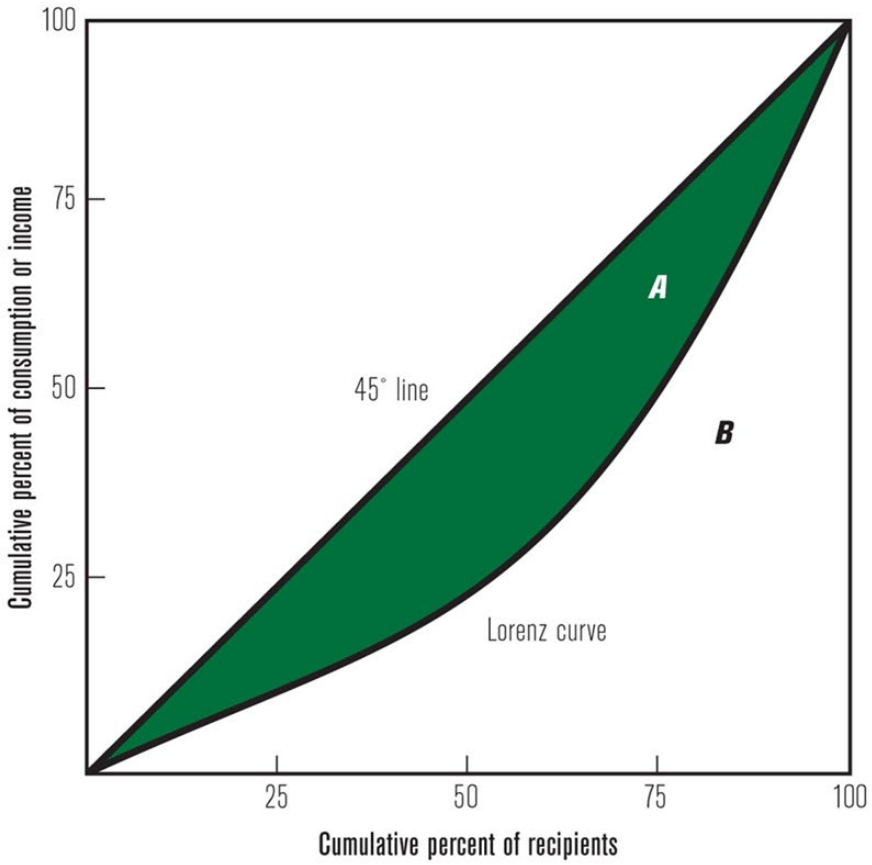


FIGURE 6-2 Lorenz Curve

Some Gini-coefficient issues (goes for most measures)

- Information is lost going from the distribution to the single measure
- Intersecting Lorenz curves can cause difficulties if only looking at the Gini coefficients. Lorenz consistent summary statistics can give conflicting results when the curves cross
- A decreasing Gini coefficient is not necessarily to the benefit of the poor (redistribution from the rich to the middle income will decrease the Gini)
- The Gini is not decomposable (sub-populations)
- Remember the absolute dimension as well – equally poor is not a desirable outcome

From graphics to a summary statistic

The Kuznets ratio: Income of x% richest/Income of y% poorest

- Does not include the whole distribution (often used by Piketty in some form)
- Fails to satisfy the so-called Dalton principle

The Gini Coefficient: $G = (1/2n^2)*\sum\sum|y_j-y_k|/average(y)$

- Includes the whole distribution – good for between country comparisons
- Satisfies the necessary fundamental principles: Lorenz consistent

Theil Index: $T_L = -(1/n)\sum\ln(y_j/\mu)$

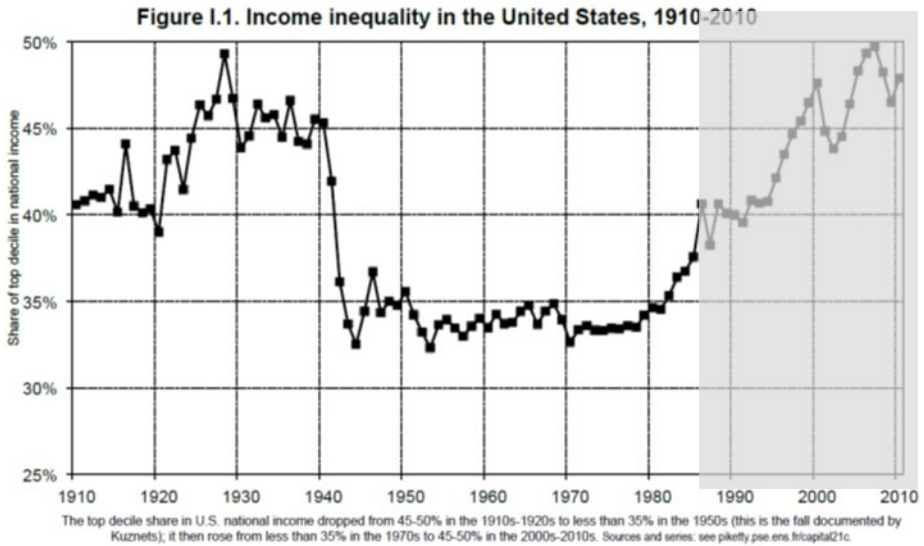
[This is the so-called mean log deviation, MLD, used by Milanovic]

- Includes the whole distribution – good for within country comparison
- Satisfies the necessary fundamental principles: Lorenz consistent

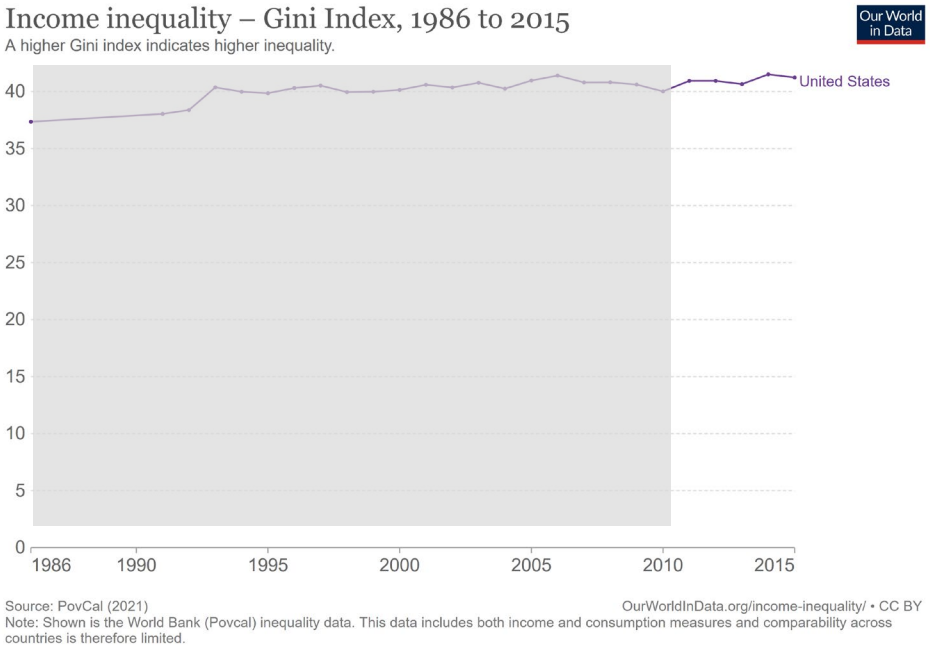
The Coefficient of Variation: $C = \sigma/\mu = [(1/n)\sum(y_j-\mu)^2]^{1/2}/\mu$

- Includes the whole distribution
- Satisfies the necessary fundamental principles: Lorenz consistent

Within Country Inequality I
Advanced economies



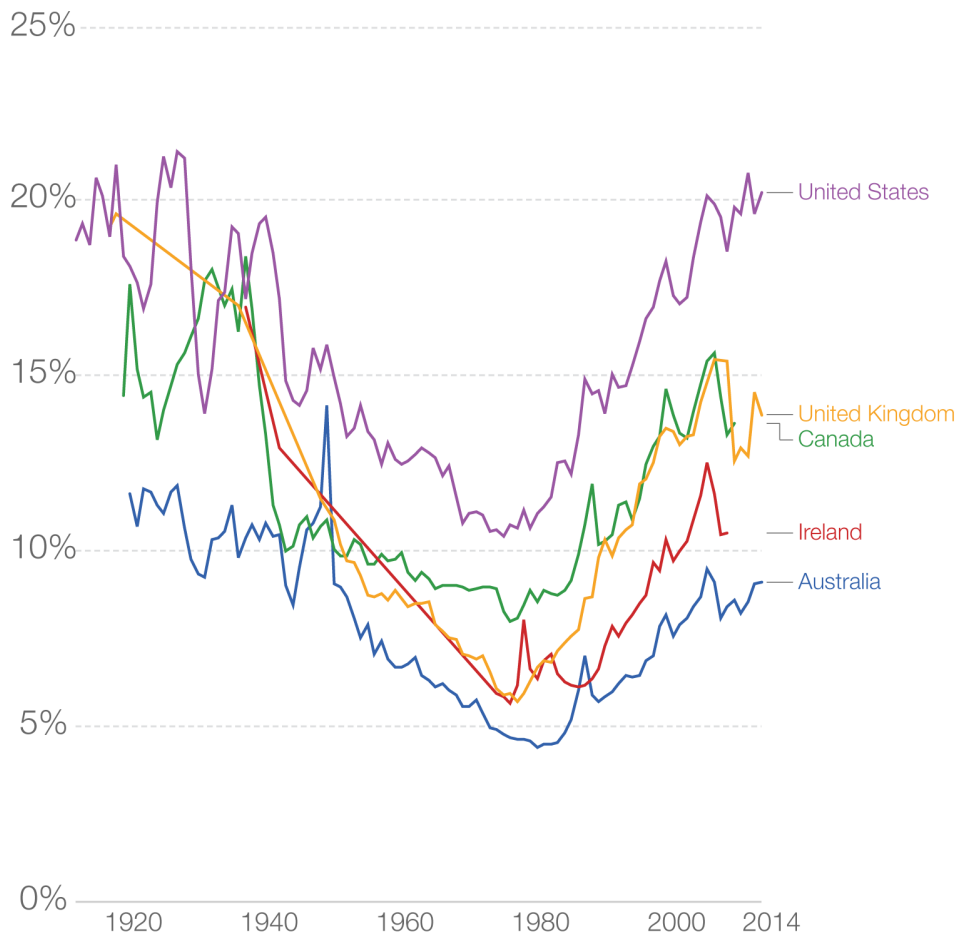
Income inequality in USA 1910-2011
Source: Piketty (2014)



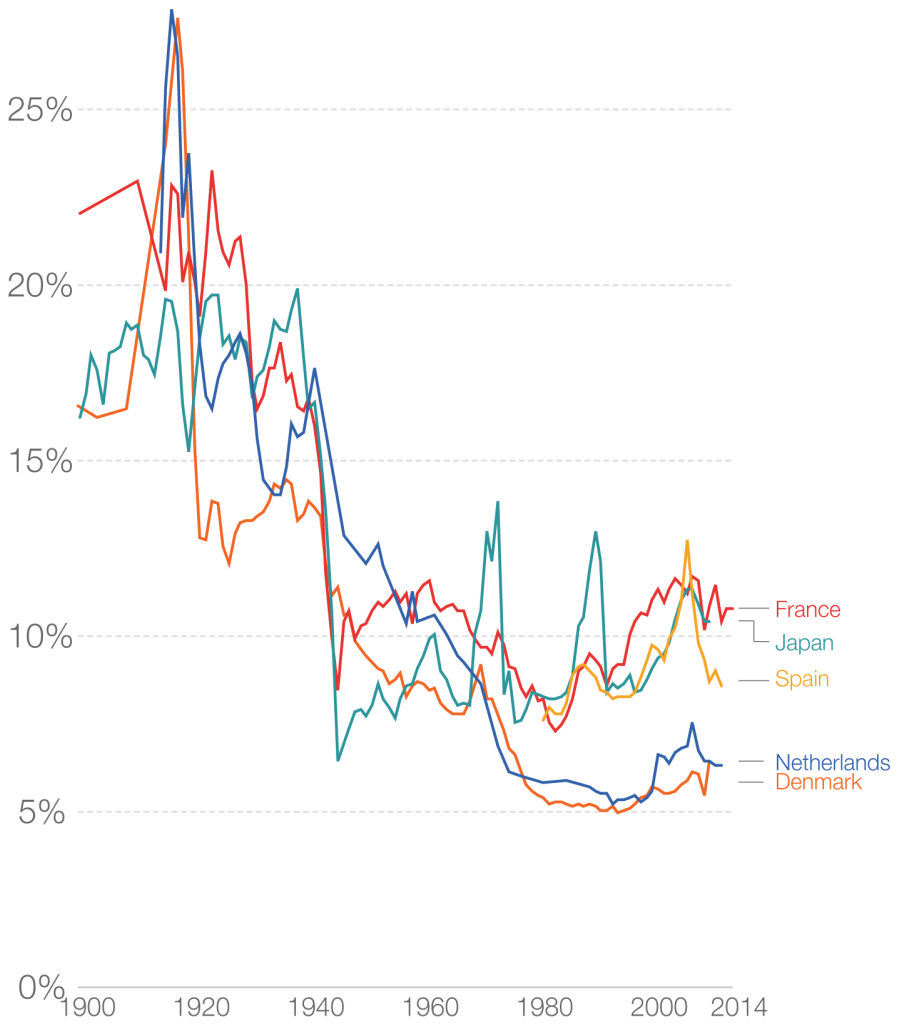
Inequality in USA measured by the Gini-coefficient and the share of the top decile in national income

Share of Total Income going to the Top 1% since 1900

The evolution of inequality in English speaking countries followed a U-shape



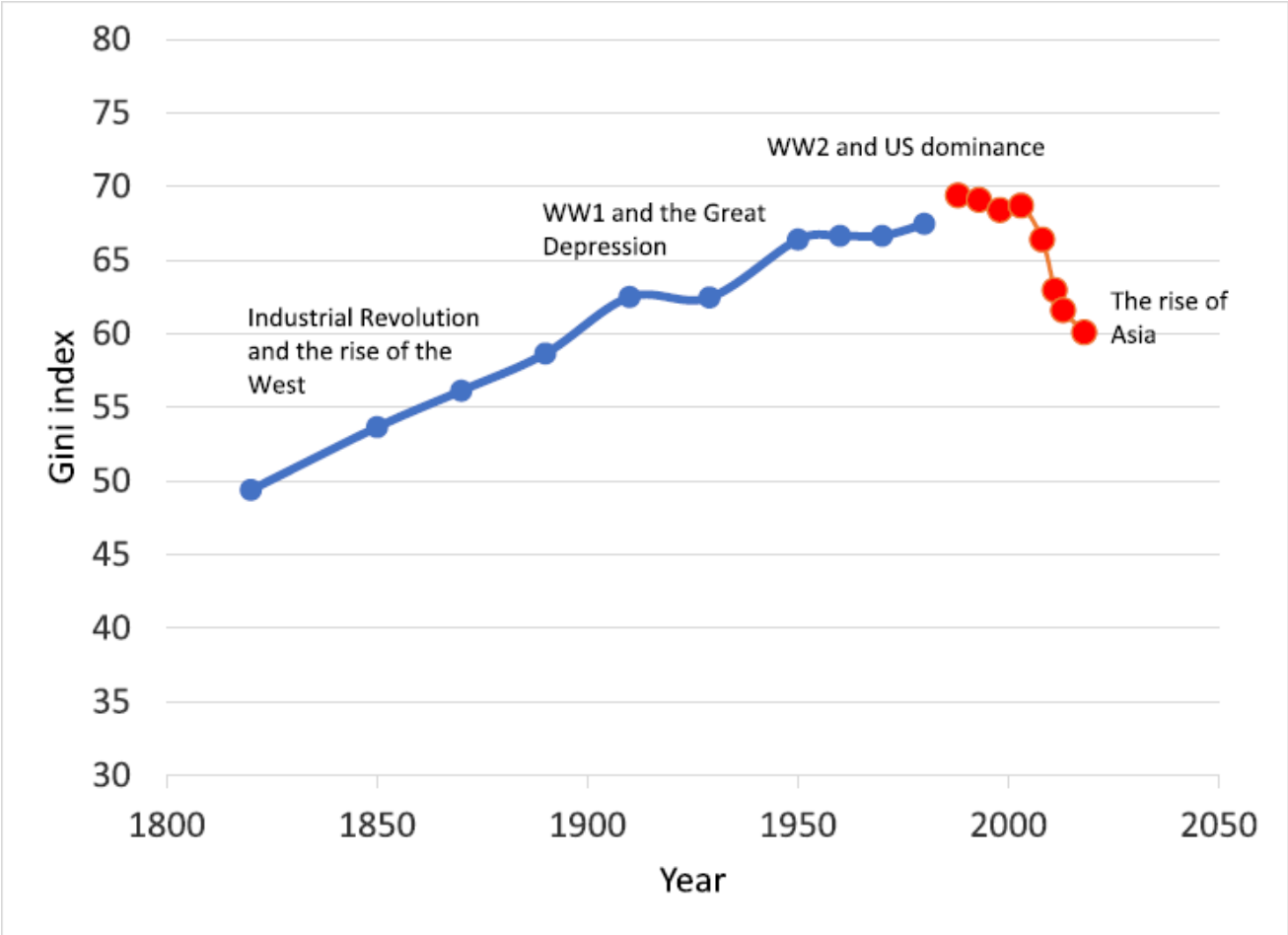
The evolution of inequality in continental Europe and Japan followed an L-shape



Data source: World Wealth and Income Database (2018). This is income before taxes and transfers.

This data visualisation is available at OurWorldinData.org. There you find the raw data and more visualisations on inequality and how the world is changing. Licensed under CC-BY-SA by the author Max Roser.

Global Inequality (Milanovic)

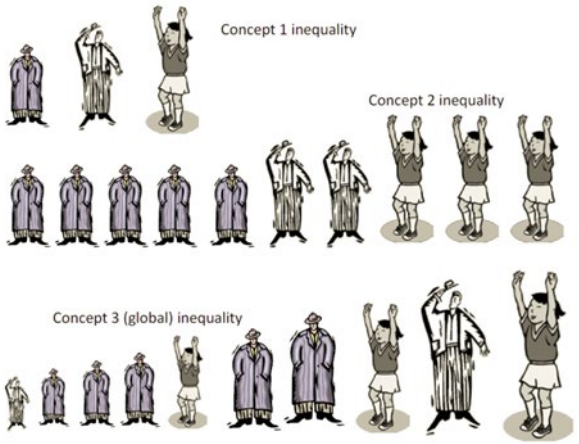


First era 1820–1950: Steadily rising global inequality.

Second era 1950-2000: Period of very high global inequality

Third era 2000-now: Global inequality is decreasing The decrease, having occurred over less than 20 years is very sharp.

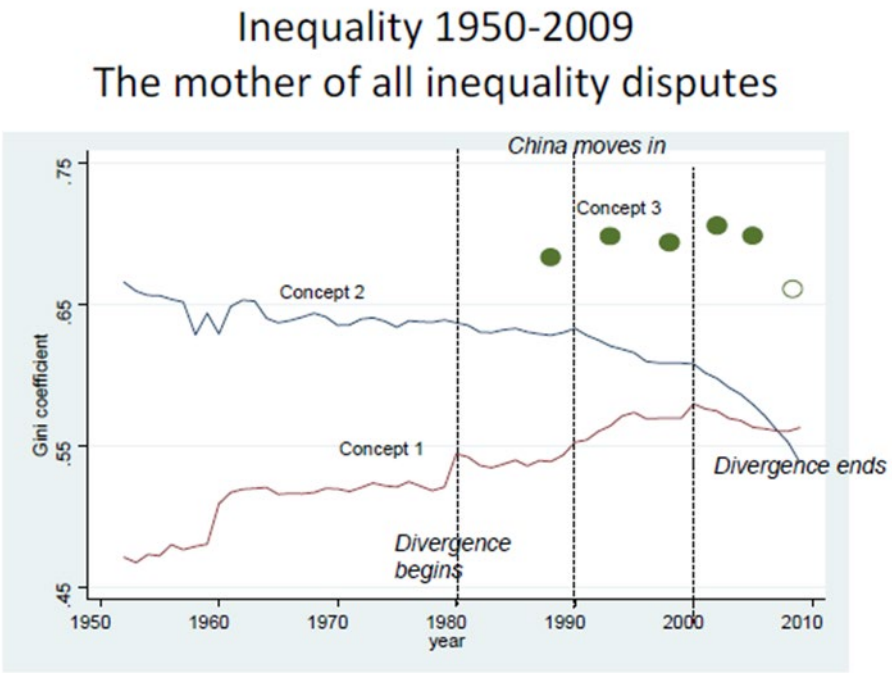
Cross-country and global inequality



Concept 1 is the cross-country Gini-coefficient based on *income per capita* across countries, **without** population weighting

Concept 2 is the cross-country Gini-coefficient based on *income per capita* across countries, **with** population weighting

Concept 3 is based on *individual incomes*



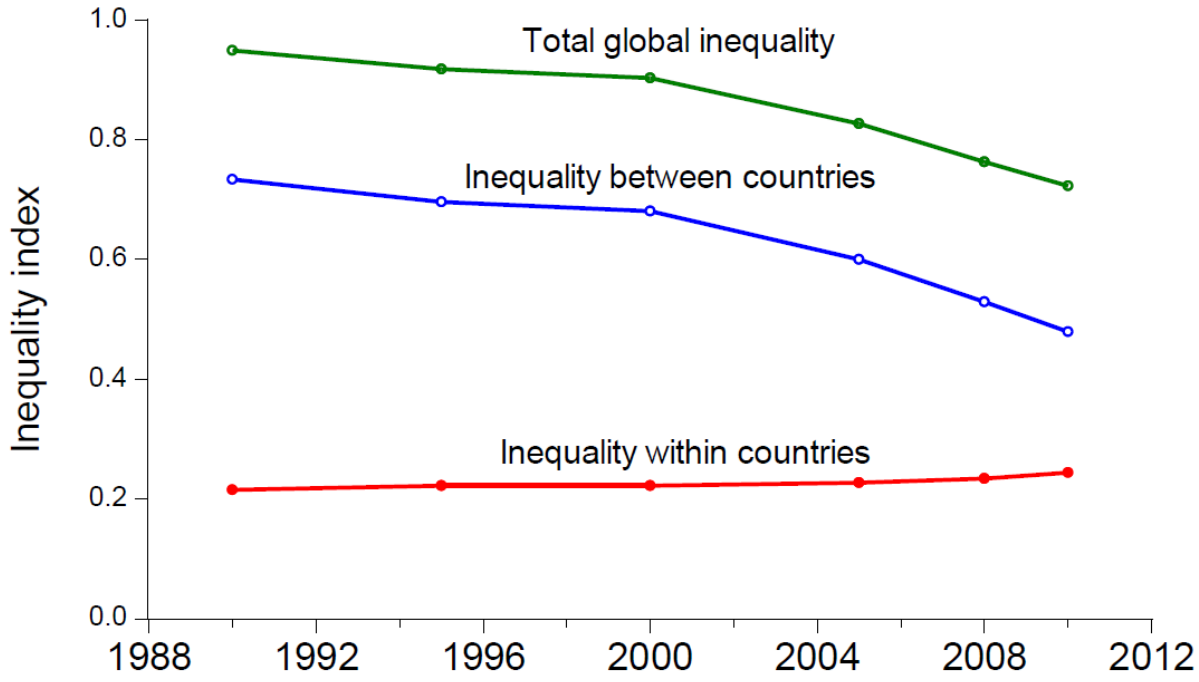
With new PPPs

- About 8 percent of the world population receives one half of global income
- Global inequality showed no clear trend, despite the development in China and India
- This is almost certainly the highest level of relative and absolute global inequality at any point in human history

Decomposition of global inequality (using definition 3)

- Slight increase in inequality **within countries** (red curve)
- Decrease in inequality **between countries** (blue curve)
- Net effect:
  Decrease in global inequality
  (Based on concept 3) (green curve)

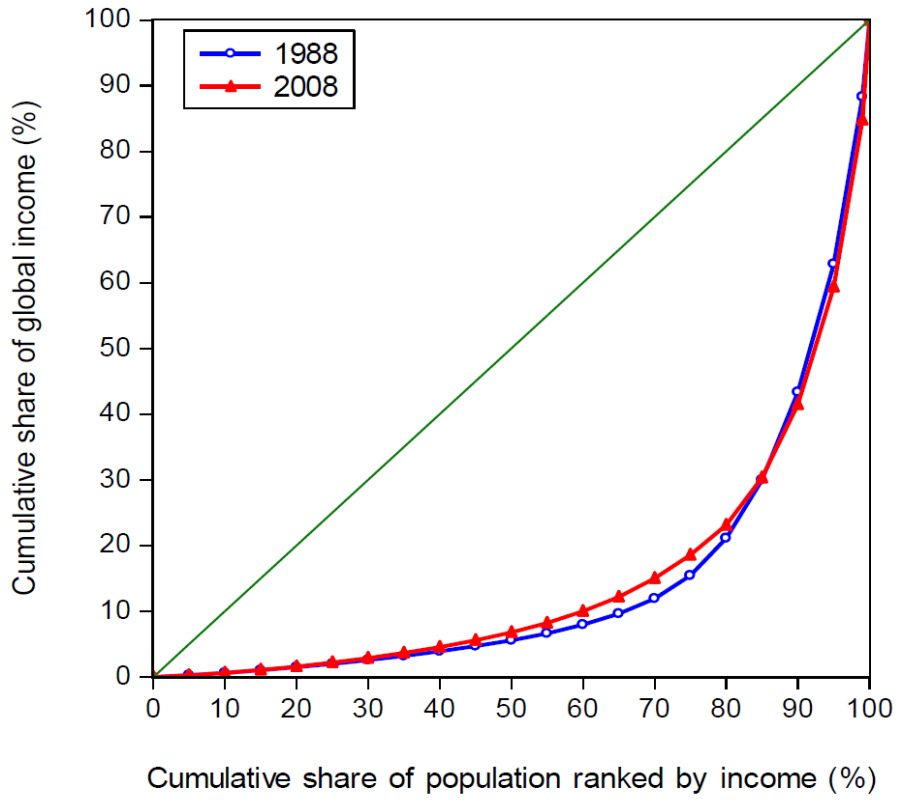
Figure 1: Global inequality and its between- and within-country components



Source: Author's illustration based on Bourguignon (2016: table 1).

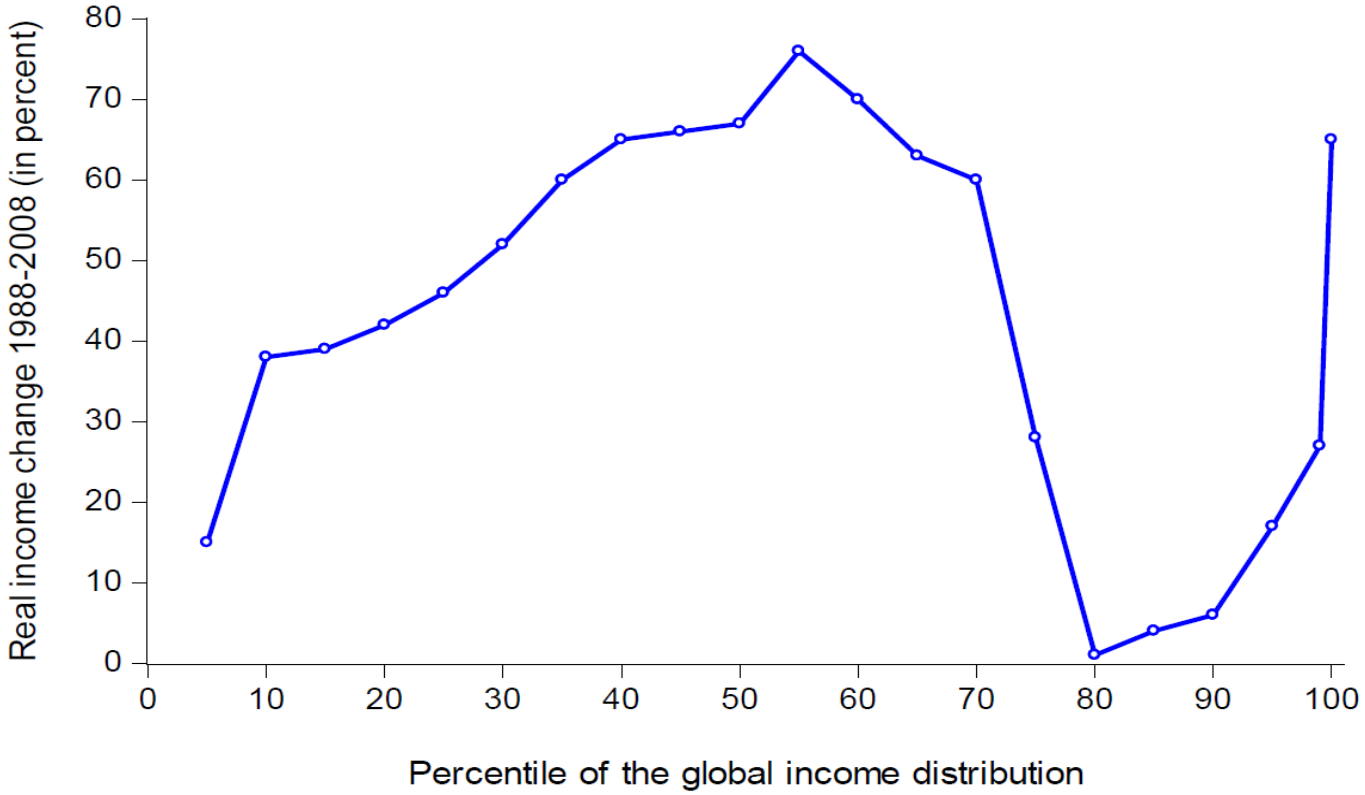
What can a decrease in inequality look like? The Elephant Graph

z curves for global income 1988 and 2008



's illustration based on estimates in Lakner and Milanovic (2016a).

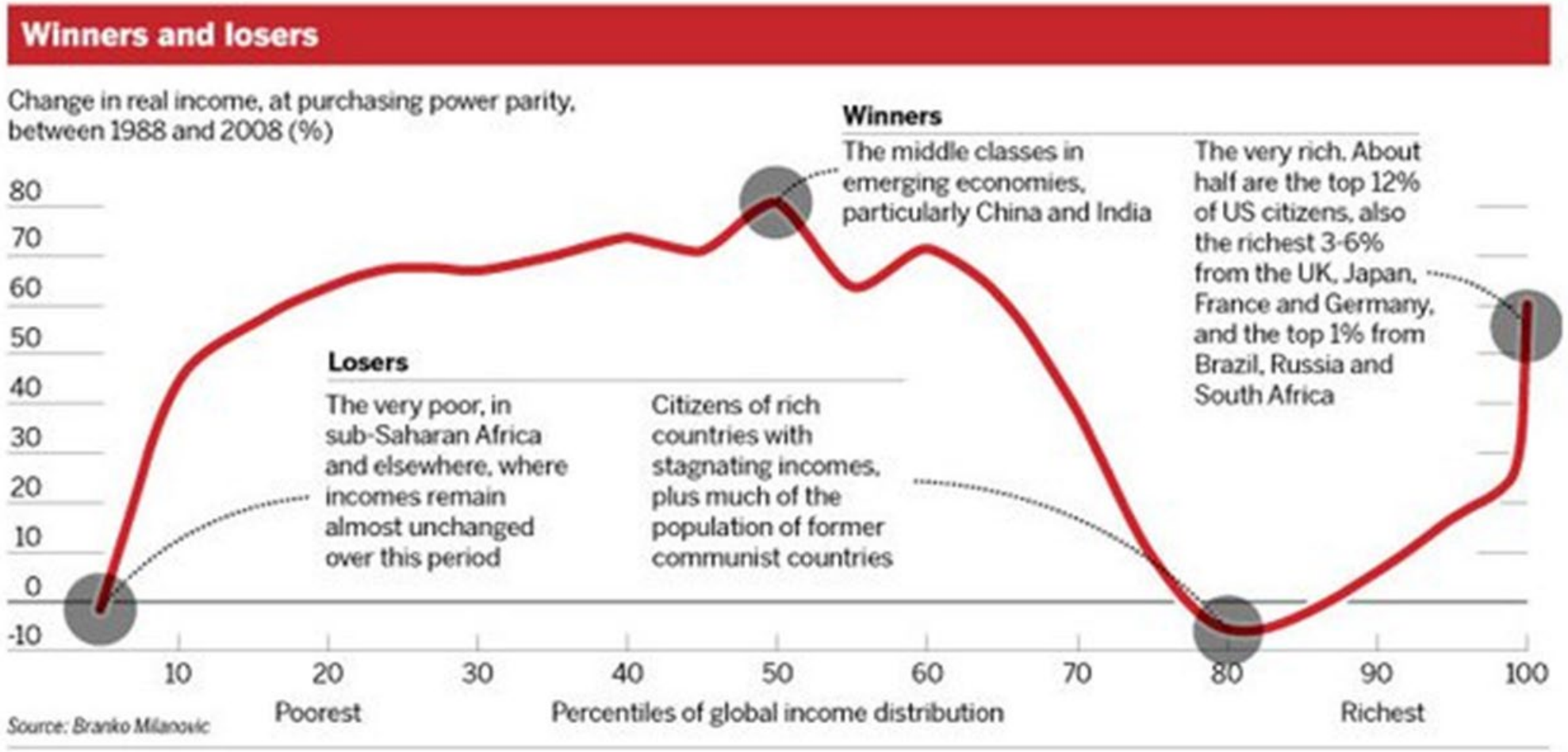
The elephant graph of Lakner and Milanovic



Author's illustration based on estimates in Lakner and Milanovic (2016a).

Note: When Lorenz curves cross then the different summary statistics may give different results in term of increases and decreases

Milanovic's Inequality Elephant: Evolution in global incomes

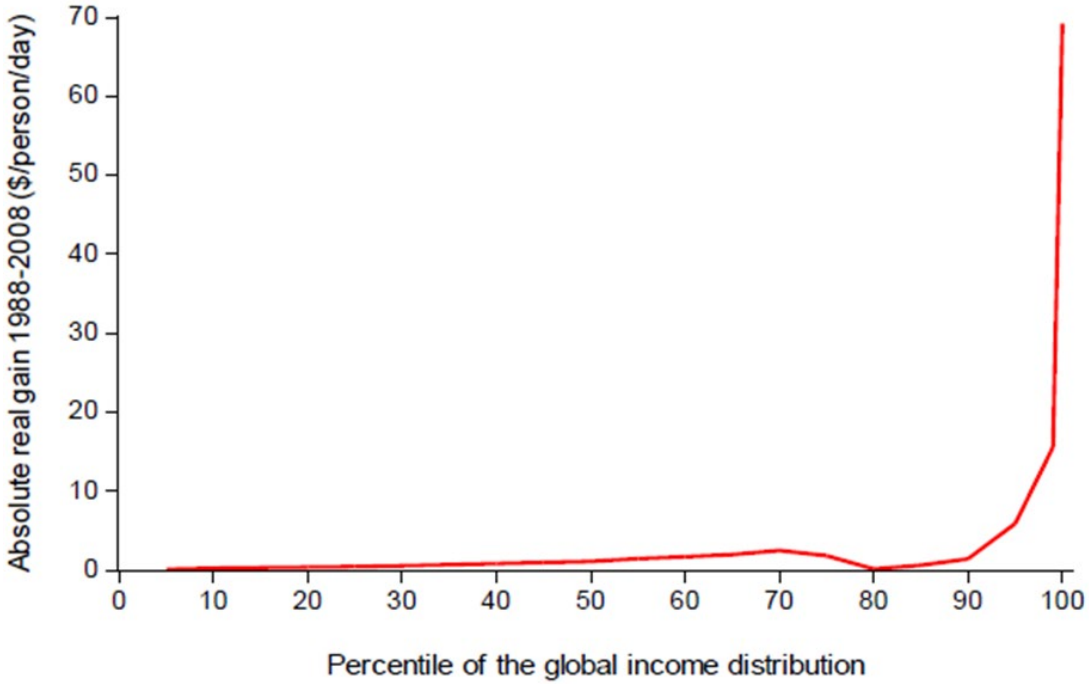


Two plots of the same data: Relative and absolute changes in incomes across the distribution

The elephant graph of Lakner and Milanovic

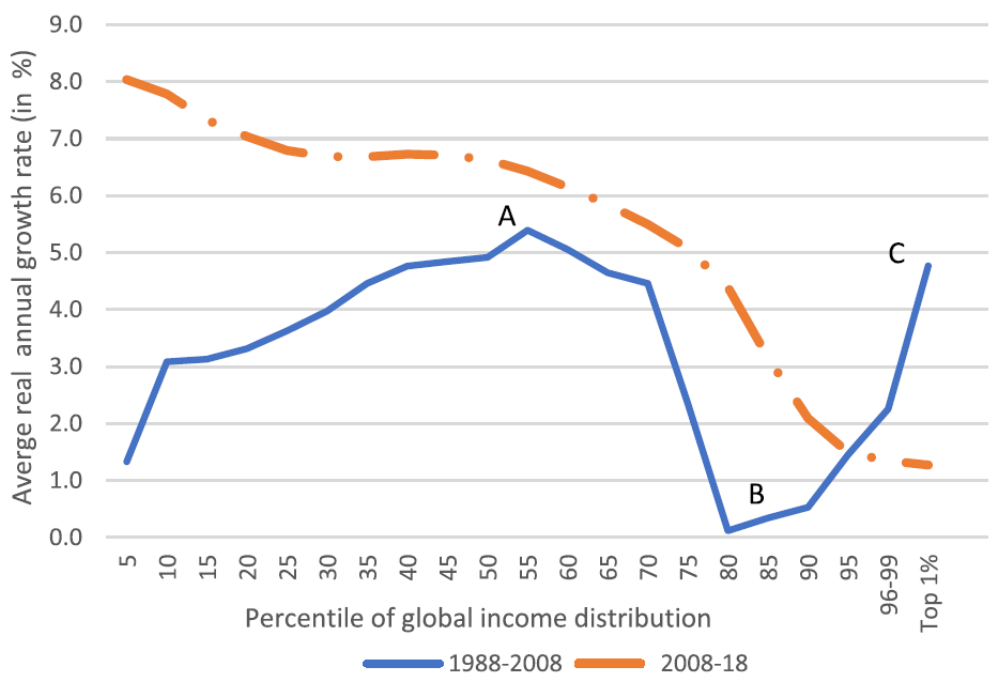
Author's illustration based on estimates in Lakner and Milanovic (2016a).

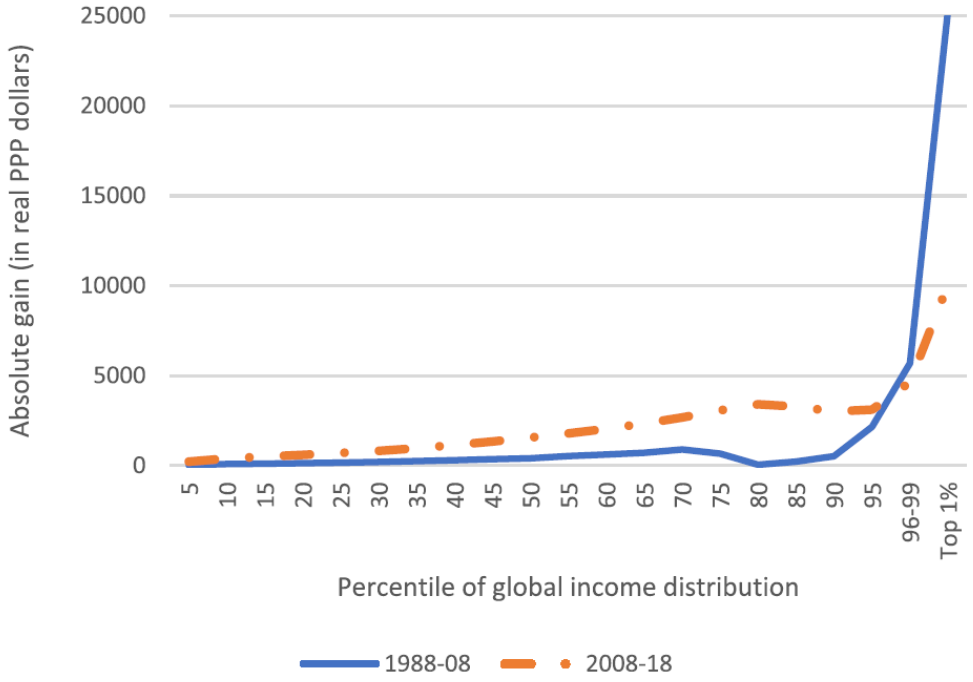
Elephant or serpent? Incidence of the absolute income gains corresponding to Figure 5



There has been a sharp increase in **absolute inequality**

The Elephant graph – Is it still relevant?





Is there a poverty-inequality trade off?

- Ravallion (2005): Are we seeing falling poverty with rising inequality, and rising poverty with falling inequality?
- Defining concepts: What is poverty and inequality?
- Ravallion only considers **absolute poverty**: poverty assessed relative to poverty lines in local currencies that attempt to have fixed purchasing power over commodities across countries.
- **Relative poverty**: poverty line rises with average income (not considered in the paper)

Relative vs. Absolute Inequality

- **Relative inequality**: Relative inequality is a function of the ratios of individual incomes to the mean.
  - Implication: If all incomes are multiplied by a constant then inequality is unchanged.
- **Absolute inequality**: depends on the absolute differences in levels of living, rather than relative differences
  - Implication: Absolute inequality is unchanged if all incomes increase by the same amount.
- Example:
  - HH1: 1.000 DKK to 2.000 DKK
  - HH2: 10.000 DKK to 20.000 DKK
  - The rich household remains 10 times richer – relative inequality unchanged.
  - Absolute difference has doubled (from 9.000 to 18.000) – absolute inequality has increased.

Results summary – concepts important

- Many people think about "inequality" in absolute terms.

- **Concepts are important**: Whether one thinks about inequality in relative or absolute terms is crucial to the position one takes on the trade off between poverty and inequality.

- Conclusion of the analysis:
  - No trade-off between absolute poverty and relative inequality.
  - Trade-off between absolute poverty and absolute inequality.

The Ravallion "Trick"

- **Question**: Is there an unconditional correlation between poverty and (absolute and relation) inequality?
- **The Ravallion "trick"**: Study the changes over time in both measures of poverty and inequality.
  - Assumption: controlling for "fixed effects" will adequately sweep away any extraneous other factors at country level.
- **Measures**:
  - Relative inequality – Gini-index
  - Absolute inequality – Gini-index controlling for mean income or consumption (the absolute Gini-index)

Understanding our inequality measures in detail

- Figure: Lorenz curve
- Relative Gini (G)= A/(A+B)
- Absolute Gini (G*) = Relative Gini * mean income or consumption
- The Ravallion "trick"

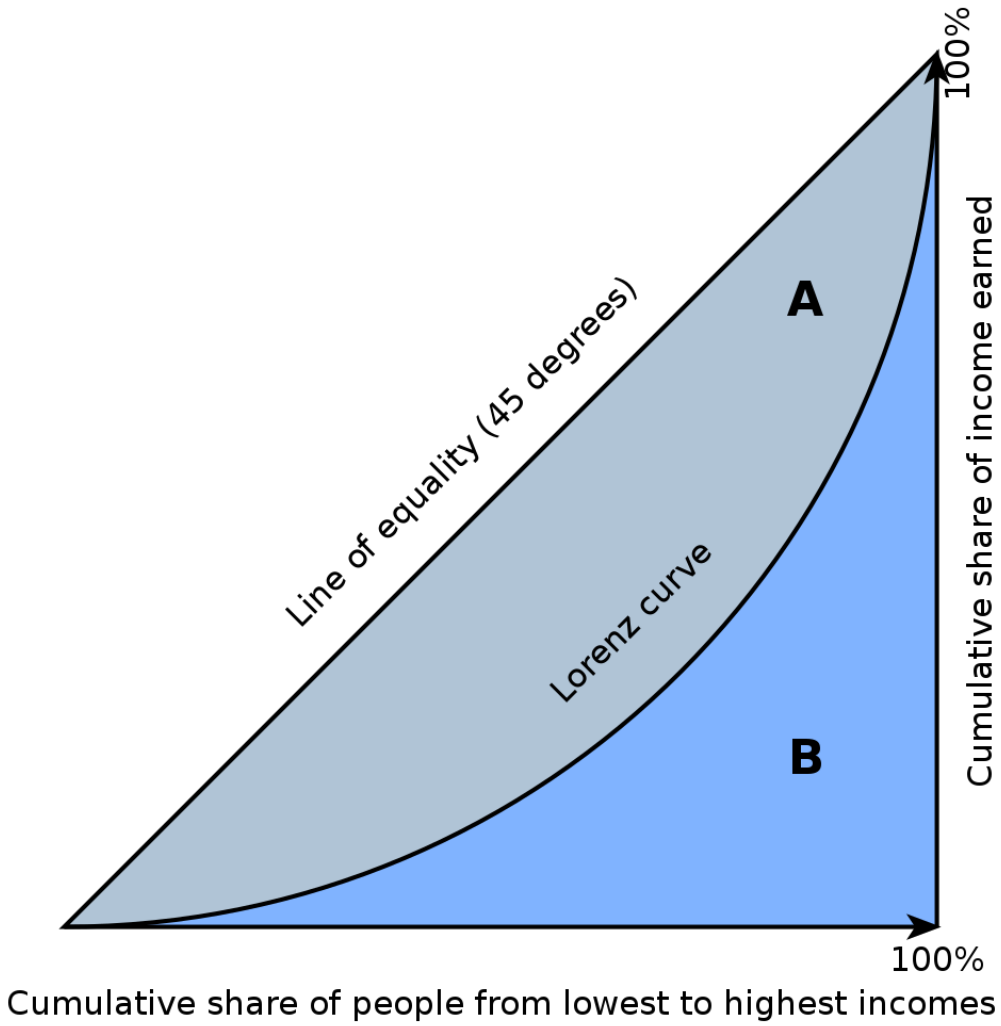
$$G^* = G * \mu \Rightarrow lnG^* = lnG + ln\mu \Rightarrow$$

$$\Delta lnG^* = \Delta lnG + \Delta ln\mu$$

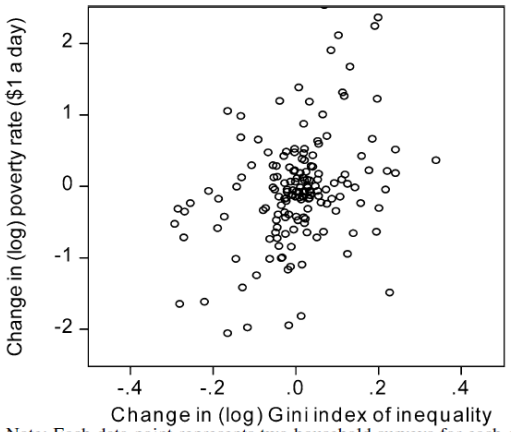
- Regressions (suppression the constant and error terms):

Relative: $\Delta lnP = \beta \Delta lnG$

Absolute: $\Delta lnP = \beta(\Delta lnG + \Delta ln\mu)$



Results overview – Ravallion (2005)



Note: Each data point represents two household surveys for each country, with multiple observations over time for most countries. The calculations were done from the World Bank's *PovcalNet* data tool, which provides income and expenditure distributions from about 400 household surveys; see http://iresearch.worldbank.org/povcalnet. The international poverty line is $1.08 a day at 1993 Purchasing Power Parity. For further discussion see Chen and Ravallion [5].

Figure 1. Change in poverty plotted against changes in relative inequality.

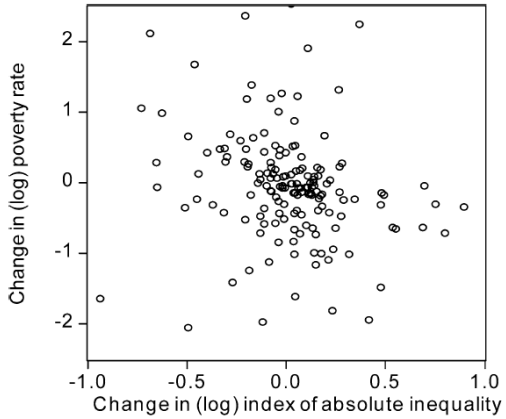


Figure 2. Change in poverty plotted against changes in absolute inequality.

<u>Relative inequality</u>:
Positive correlation
Growth splits – does not change the result.
No support for the idea of the KH (a PIT for relative inequality).

<u>Absolute inequality</u>:
Negative correlation
Growth splits – some indication of differences.
Some support for the idea of the KH (a PIT for absolute inequality).

If relative inequality does not change on average with growth then absolute inequality will tend to rise with growth.
Thus, we see a negative relationship between changes in poverty and changes in absolute inequality.

Conclusion: Poverty-Inequality Relationship

- No sign of a systematic trade off between **absolute poverty** incidence and **relative inequality**.
  - Lower absolute poverty tends to come hand in hand with lower relative inequality
  - Reason: economic growth shows little correlation with changes in relative inequality

- Strong sign of a systematic trade off between **absolute poverty** incidence and **absolute inequality**.
  - Lower absolute poverty is associated with higher absolute inequality

- We will look closer at this relationship during the exercise using updated data.

Browsing the data

Poverty and inequality data https://pip.worldbank.org/poverty-calculator

- Go to poverty calculator and select "World" – what is the Global Poverty difference between having a 1.9$ and 3.2$ benchmark?
- Select Gini Index – browse for example Bolivia, Tanzania and Nepal and go back and forth between the Gini and the Poverty Rate. Try and change to Brazil, Ethiopia and Vietnam.
- Browse country profiles on Poverty and Inequality Platform, select for example Bolivia, Tanzania and Nepal and then Brazil, Ethiopia and Vietnam
- Browse "other measures" (for example Multidimensional Poverty) (https://pip.worldbank.org/multidimensionalpoverty-measure)